XO Platform

TECHNICAL GUIDE

Payment Integration

The technical and business steps for a real integration with Paymob, mobile wallets, Fawry and InstaPay, and what the platform already has in place today.

Paymob · Cards | Mobile wallets | Fawry · FawryPay | InstaPay | HMAC · Webhooks

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01

THE BIG PICTURE

Aggregator or direct integration?

There are two ways to accept electronic payments in Egypt: contract with one payment aggregator that covers most methods, or contract with each company separately.

APPROACH	WHAT IT MEANS	ASSESSMENT
Aggregator (e.g. Paymob)	One contract and one API covering cards, wallets and cash payment at outlets.	Recommended The current code is built on it.
Direct integration	A separate contract with each company (Fawry, Vodafone, banks), each with its own API, signature scheme and reconciliation.	Only when needed Workable for Fawry. Vodafone and banks need business partnerships and offer no public merchant APIs.

Recommendation

One aggregator (Paymob) for cards, wallets and cash; add direct Fawry, or InstaPay through an aggregator that supports it, only if nurseries need them.

Payment methods and who covers them

METHOD	THROUGH PAYMOB	DIRECT
Cards (Visa · Mastercard · Meeza)	One integration; 3-D Secure (OTP) runs on Paymob's page.	Not practical (needs a card-acquiring licence from a bank).
Mobile wallets	One integration covering Vodafone Cash, Etisalat Cash, Orange Cash, WE Pay and bank wallets.	A business agreement with each operator; no public API.
Cash at outlets	The Kiosk channel (Aman and Masary networks).	FawryPay, for Fawry codes specifically.
InstaPay	Confirm with the account manager whether it is available on your account.	Through a bank or an aggregator that supports the Instant Payment Network (e.g. Kashier).

BEFORE ANYTHING ELSE

02 Architecture decision: where does the money go?

The current code reads the Paymob keys from global server secrets (such as `PAYMOB_SECRET_KEY`), so **every nursery's money currently goes to a single merchant account**. This must be settled before going live.

OPTION 1 · RECOMMENDED

A Paymob account per nursery

- Each nursery opens a merchant account in its own name and bank account.
- Each nursery's keys are stored encrypted (e.g. Supabase Vault).
- The server picks the keys by `invoice.nursery_id`.
- Money lands directly in the nursery's account, with no intermediate settlement.

OPTION 2

The platform collects on the nurseries' behalf

- Needs a main-merchant / sub-merchant arrangement with Paymob.
- Needs a legal status as a payment facilitator under Central Bank of Egypt oversight.
- The platform is responsible for distributing and reconciling the money per nursery.
- Much more complex, technically and legally.

Code change needed for option 1

A per-nursery key table or store, and a lookup that picks the keys by nursery in `invoice-checkout`, `parent-signup-complete` and `paymob-webhook` (an HMAC secret per nursery).

03

PAYMOB

Cards, wallets and cash

Business requirements

- **Merchant account:** commercial register, tax card, and a bank account for settlements (merchant KYC).
- **Website pages:** terms and conditions, refund policy, privacy policy and contact details. Gateways review them before activation.

Payment flow

1 · Parent
Chooses "Pay online"

→

2 · Our server
Creates a payment attempt and an Intention

→

3 · Paymob
Unified checkout: card / wallet / code

→

4 · Webhook
HMAC-signed result reaches the server

→

5 · Settlement
Confirmed, failed or pending code, then the result page

Technical steps

#	STEP	STATUS IN THE PROJECT
1	Sandbox account: `secret` key, `public` key and test integration IDs.	**Needed from you**
2	Create an **Integration** per payment method in the Paymob dashboard, each with its own ID. A test ID works only with the test key, and a live ID only with the live key.	**Needed from you**
3	Register the notification URL (`paymob-webhook`) on every integration and copy the HMAC secret.	**Needed from you · code ready**
4	The server creates an **Intention**: amount in piasters, `special_reference` set to the payment attempt ID, plus `notification_url` and `redirection_url` .	**Ready**
5	Redirect the parent to the unified checkout; card OTP and wallet PIN happen there only.	**Ready**
6	Receive the result: verify the signature, match amount and currency, then confirm, fail or keep the payment pending.	**Ready**
7	Refund and void through the Paymob API, to handle double payments.	**Not built**
8	Daily reconciliation between Paymob transactions and the `payments` table.	**Not built**
9	Saved cards: store the token Paymob sends in a `TOKEN` callback, never the card itself.	**Optional · not built**

Wallets through Paymob

- One integration covers Vodafone Cash, Etisalat Cash, Orange Cash, WE Pay and bank wallets.
- The parent enters their wallet number on Paymob's page and confirms with an OTP or PIN on the wallet side.
- Our site never sees the PIN.
- A direct Vodafone Cash integration needs a business agreement and has no public API.

Cash through Paymob

- Paymob's documented cash channel is **Kiosk**, through the Aman and Masary networks, **not Fawry**.
- The code has a variable named `PAYMOB_FAWRY_INTEGRATION_ID`.
- Ask Paymob whether Fawry codes are available on your account; if not, that ID is for Kiosk and its name and the UI texts should change.

Server secrets needed to go live

SECRET	VALUE
`PAYMOB_SECRET_KEY` · `PAYMOB_PUBLIC_KEY`	From the Paymob dashboard (test first, then live).
`PAYMOB_CARD_INTEGRATION_ID`	The cards integration ID.
`PAYMOB_MOBILE_WALLET_INTEGRATION_ID`	The wallets integration ID.
`PAYMOB_INSTAPAY_INTEGRATION_ID` · `PAYMOB_FAWRY_INTEGRATION_ID`	When available on the account (see the Kiosk note).
`PAYMOB_AMOUNT_MULTIPLIER`	**100** (the amount is sent in piasters).
`PAYMOB_HMAC_SECRET`	**Required**: without it, online payment is not offered at all.
`APP_BASE_URL`	The public site URL over HTTPS.

FAWRY

04 Direct integration through FawryPay

If a "Pay at Fawry" code is specifically required, the route is a direct merchant account with Fawry.

1. **Fawry merchant account:** you get a `merchantCode`, a `secureKey` and a staging environment.
2. **Issue the code:** `POST /ECommerceWeb/Fawry/payments/charge` with `paymentMethod: PayAtFawry`. The response carries `referenceNumber` (the code) and `expirationTime`.
3. **Show the code:** large type, the expiry date, a copy button and an SMS button (through Fawry or our existing `sms-dispatch` function). The code screen exists in the platform; it only lacks the expiry date.
4. **Payment notification:** when the parent pays at an outlet, Fawry's server sends a signed notification to a new `fawry-webhook` function; we verify the signature, then confirm the payment.
5. **Status inquiry:** for daily reconciliation and for expired codes.

Charge request signature (SHA-256)

```
signature = SHA256( merchantCode + merchantRefNum + customerProfileId
                    + paymentMethod + amount(2 decimals) + secureKey )
```

`merchantRefNum` is our payment attempt ID; `customerProfileId` is left empty when there is none. The signature scheme of the payment notification must be taken from Fawry's official documentation; it was not published in the public sources.

Code change needed

Generalise `paymob_settle_attempt` into `gateway_settle_attempt(provider, …)`. The `provider` column already exists on `payment_attempts`, and `payment_events` supports more than one provider. Add two functions: `fawry-charge` and `fawry-webhook`.

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INSTAPAY

Request to Pay and QR through an aggregator

There is no public merchant API. The Instant Payment Network (IPN) is owned by the Central Bank of Egypt; access is through a bank or payment company that supports **Request to Pay** or **QR** payments.

DOCUMENTED EXAMPLE

Kashier

- An order with `INITIATE_R2P` and the parent's InstaPay address (`payerAddress`), or `GENERATE_QR` .
- The parent approves in the InstaPay app with their IPN PIN.
- The result reaches the server through a webhook.
- No saved cards and no recurring payments.

PAYMOB

Ask the account manager

Confirm whether Paymob offers InstaPay this way on your account. If it does, adding `PAYMOB_INSTAPAY_INTEGRATION_ID` is enough and the current flow works.

What we need to build

- A field for the InstaPay address (such as `name@instapay`) or the registered mobile number.
- A waiting screen with a countdown that follows the payment status live (Realtime on `payment_attempts`) and switches to "Payment successful" as soon as the notification arrives.
- The current method (manual transfer, then the transaction number) stays as a fallback.

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FIXED RULES

Security and compliance

Card data

Never passes through our server. The hosted checkout keeps PCI-DSS scope at its simplest level (SAQ A). No card number, CVV, PIN or OTP fields on our pages.

Verify every result

The signature (HMAC-SHA512 for Paymob, SHA-256 for Fawry), amount and currency match, and duplicate protection. Every result is stored in `payment_events`.

Secrets and logs

Secrets live in Supabase Secrets or Vault only; logs never contain card numbers or full national IDs.

Kill switch

A setting that turns off online payment immediately during an incident. Listed in the roadmap document; not built yet.

Already built in the platform

- Paymob results are applied automatically after signature verification: confirmed for the exact amount only, failed with the reason, or a pending code.
- A payment-failed screen (reason, try again, change method, contact the nursery), a payment code screen, and an "I didn't complete the payment" button.
- A mismatched amount, or money for a rejected attempt, is never applied automatically; it goes to finance for review.
- Online payment is not offered at all without `PAYMOB_HMAC_SECRET`.

07

DELIVERY

Suggested implementation plan

1. **Business decision:** a Paymob account per nursery or one account for the platform, then start merchant KYC.
2. **Testing:** put the sandbox keys on the staging project and try every case: success, decline, OTP failure, leaving without paying, and a pending code.
3. **Per-nursery keys:** implement them in the code if option 1 is chosen.
4. **Go live** with cards and wallets, then make one small real payment and refund it.
5. **Refunds and daily reconciliation:** build refunds through the Paymob API and a daily comparison with the `payments` table.
6. **Direct Fawry and InstaPay** through an aggregator that supports them, as nurseries need them.

Go-live checklist

Live keys and matching live integration IDs · `PAYMOB_AMOUNT_MULTIPLIER = 100` · `PAYMOB_HMAC_SECRET` · notification URL registered on every integration · `invoice-checkout` and `paymob-webhook` deployed · methods enabled in nursery settings · one small real payment, then refund it.

08

REFERENCES

Sources

Documentation sources used in this guide. Unofficial sources (community guides and libraries) must be checked against the provider's official documentation before relying on them.

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